SF2940 Probability: Exam

Fall 2023

Date: 19. December 2023, 08:00-12:00.
Examiner: Kevin Schnelli, SCI MAT. Phone: 72 02
Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Other: Solutions will be posted on Canvas within a week after the exam.

Problem 1. Let $X_1, X_2, X_3, \dots$ be a sequence of random variables, let X be another random variable.

a.) Define what it means that X_n converges in distribution to X, as $n \to \infty$.
..... (2points)

b.) Assume now in addition that $X_1, X_2, X_3, \dots$ satisfy

$$\mathbb{P}(X_n = 0) = 1 - \frac{1}{n}\,, \qquad \mathbb{P}(X_n = 2) = \frac{1}{2n}\,, \qquad \mathbb{P}(X_n = -2) = \frac{1}{2n}\,,$$

for each $n = 1, 2, 3, \dots$. Show that X_n converges in distribution as $n \to \infty$ and determine the limit. ..... (1point)

c.) Assume further that $X_1, X_2, X_3, \dots$ are all independent. Set now

$$Y_n := \sum_{k=1}^{n} \frac{X_k}{\sqrt{k}}\,.$$

Compute $\lim_{n\to\infty} \mathbb{E}[Y_n^2]$..... (2points)
Hint: Recall that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}\,.$$

Problem 2. The bivariate random variable (X, Y) has the joint probability density

$$f_{X,Y}(x,y) = \begin{cases} 2xy\mathrm{e}^{-x-y}\,, & \text{if } y > x > 0\,, \\ 0\,, & \text{else}\,. \end{cases}$$

a.) Find the joint probability density of (U, V) where

$$U := \frac{X}{X+Y}\,, \qquad V := X + Y\,.$$

..... (3points)

b.) Compute the marginal densities of U and V, and determine whether U and V are independent or not. ..... (2points)

Problem 3. Let $X \in \mathrm{N}(0,1)$ be a standard Gaussian random variable. Let $U \in \mathrm{Be}(\frac{1}{2})$ be a Bernoulli random variable that is independent of X.

Introduce the random variable Y by setting

$$Y := \begin{cases} X\,, & \text{if } U = 0\,, \\ -X\,, & \text{if } U = 1\,. \end{cases}$$

a.) Determine the distribution of the random variable Y..... (2points)

b.) Determine the covariance matrix of the random vector $(X, Y)'$..... (1point)

c.) Is the random vector $(X, Y)'$ a bivariate Gaussian vector, or not? Justify your answer carefully. ..... (2points).

Problem 4. Let $X_1, X_2, X_3, \ldots$ be a sequence of i.i.d. random variables with mean $\mathbb{E}X_1 = \mu > 0$ and finite variance. Let $N_1, N_2, N_3 \ldots$ be another sequence of random variables that are independent of $(X_i)_{i=1}^{\infty}$ such that, for each $n = 1, 2, 3 \ldots$, $N_n \in \mathrm{Ge}(p_n)$, where $p_n \in (0, 1)$. Assume further that $p_n \to 0$, as $n \to \infty$.

a.) Show that $p_n N_n$ converges in distribution, as $n \to \infty$, and determine the limit. ..... (3points)

b.) Consider now the random sum of random variables

$$S_{N_n} := X_1 + X_2 + \ldots + X_{N_n} ,$$

with the convention $S_0 = 0$.

Show that $p_n S_{N_n}$ converges in distribution to a random variable that has exponential distribution with parameter μ, as $n \to \infty$. ..... (2points)

Good luck!

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a)$, $a \in \mathbb{R}$	$p(a) = 1$	$\mathrm{e}^{\mathrm{i}ta}$
Bernoulli	$\mathrm{Be}(p)$; $0 \le p \le 1$	$p(0) = q$, $p(1) = p$; $q = 1 - p$	$q + p\mathrm{e}^{\mathrm{i}t}$
Binomial	$\mathrm{Bin}(n, p)$, $n = 1, 2, \ldots$; $0 \le p \le 1$	$p(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$(q + p\mathrm{e}^{\mathrm{i}t})^n$
Geometric	$\mathrm{Ge}(p)$, $0 < p < 1$	$p(k) = pq^k$, $k = 0, 1, 2, \ldots$; $q = 1 - p$	$\frac{p}{1 - q\mathrm{e}^{\mathrm{i}t}}$
Poisson	$\mathrm{Po}(\lambda)$, $\lambda > 0$	$p(k) = \mathrm{e}^{-\lambda} \frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$	$\mathrm{e}^{\lambda(\mathrm{e}^{\mathrm{i}t} - 1)}$

Table 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$\mathrm{U}(a, b)$, $a < b$	$f(x) = \frac{1}{b-a}$, $a < x < b$	$\frac{\mathrm{e}^{\mathrm{i}tb} - \mathrm{e}^{\mathrm{i}ta}}{\mathrm{i}t(b-a)}$
Exponential	$\mathrm{Exp}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{\lambda}\mathrm{e}^{-x/\lambda}$, $x > 0$	$\frac{1}{1 - \lambda\mathrm{i}t}$
Laplace	$\mathrm{L}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{2\lambda}\mathrm{e}^{-\lvert x\rvert/\lambda}$, $x \in \mathbb{R}$	$\frac{1}{1 + \lambda^2 t^2}$
Beta	$\beta(r, s)$, $r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1}(1-x)^{s-1}$, $0 < x < 1$	$*$
Normal	$\mathrm{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}\mu t - \frac{1}{2}\sigma^2 t^2}$
Standard normal	$\mathrm{N}(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{-\frac{1}{2}t^2}$
Log-normal	$\mathrm{LN}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(\log x - \mu)^2/\sigma^2}$, $x > 0$	$*$
Cauchy	$\mathrm{C}(m, \eta)$, $\eta > 0$, $m \in \mathbb{R}$	$f(x) = \frac{1}{\pi}\frac{\eta}{(x-m)^2 + \eta^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}mt - \eta\lvert t\rvert}$

Table 2. Continuous distributions